

Lum Wen-Shen

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SUMMARY

Software Engineering student and part-time AI Engineer with hands-on experience building production backend services, REST APIs, database-driven automation, AI integrations, background workers, containerized infrastructure, and multiplayer game systems. Experienced with Node.js, Python, SQL, Docker, LLM/RAG systems, Roblox Studio, Luau, and third-party API integrations. Available for freelance backend, AI engineering, and game-development work.

EDUCATION

Monash University Malaysia

Oct 2024 – Present

Bachelor of Engineering (Software Engineering)

CGPA: 3.73

Monash High Achiever Award

EXPERIENCE

AI Engineer (Part-time) | *Beyond Photography*

Feb 2026 – Present

- Built and deployed an AWS-hosted **WhatsApp trading platform** serving **100+ active users**, converting natural-language instructions into validated MetaTrader 5 operations through backend APIs.
- Designed a **provider-agnostic LLM integration layer** supporting model routing, structured outputs, validation, fallback handling, and external service integrations.
- Migrated legacy AI infrastructure to the **Vercel AI SDK**, simplifying the application architecture and **tripling feature deployment velocity**.
- Developed **Hidden Markov Models** for real-time market regime and volatility detection and integrated model outputs into production workflows.

AI Engineering Intern | *Beyond Photography*

Oct 2025 – Feb 2026

- Developed backend automation services using **Node.js, REST APIs, SQL, and background workers** to manage cloud-container lifecycle operations, status changes, reminders, and notifications.
- Built database-driven workflows integrating **Microsoft SQL Server, Proxmox, Nginx, and external APIs**, including automated container provisioning, shutdown, and email notifications.
- Helped build a client-facing WhatsApp AI agent and a **~90%-accuracy multimodal RAG system** with document processing and automated re-indexing.
- Optimized CI/CD and deployment workflows, reducing deployment time from **20 minutes to approximately 30 seconds**.

Head Treasurer | *Monash University Engineering Club*

Jul 2025 – Present

- Built a Python automation pipeline integrating **Asana and Google Sheets APIs** to synchronize financial records and automatically validate submissions.
- Implemented validation and compliance logic with automated alerts, reducing manual administrative work by **5+ hours weekly**.
- Led a 5-member finance committee and standardized financial review workflows, reducing processing time by **10%**.

PROJECTS

McFatty's – Roblox Restaurant-Management Game

2026 – Present

- Designed and built the core **merge, eat, refill, and sell** gameplay loop, including drag-and-drop food merging, tier progression, weight gain, restaurant income, tutorials, upgrades, perks, achievements, agents, and rebirths.
- Developed a modular **client-server architecture** supporting 20 player restaurant plots, with server-authoritative progression and economy state while clients handle responsive UI, food visuals, and NPC animation.
- Reduced unnecessary network replication by moving cosmetic animation to clients, quantizing worker-scale updates, reconciling worker state at controlled intervals, and applying distance checks and per-player rate limits to station requests.
- Created a data-driven agent and worker system using shared Luau configuration modules, runtime station assignment, rarity-based presentation, fallback character rigs, and cross-platform mouse, touch, and gamepad input.

NextHack 2026 – AI Fraud Detection Platform (Finalist)

Jul 2026

- Built a real-time fraud detection Chrome extension that analyzes transaction data before checkout execution.
- Designed a parallel **multi-agent architecture** with Risk, Behavior, and Anomaly agents producing structured outputs for downstream decision logic.
- Built backend integrations for transaction processing, AI inference, and aggregation of agent results.

Cloud Container Automation Platform

2025 – 2026

- Built Node.js services connecting **SQL Server, Proxmox, Nginx, email APIs, and background workers** to automate cloud container operations.
- Implemented database-backed status tracking, scheduled expiry handling, renewal reminders, error handling, and automated lifecycle actions.

SKILLS & LANGUAGES

- **Backend:** Node.js, Express, Python, REST APIs, background workers, API integrations
- **Databases:** SQL, Microsoft SQL Server, database schemas, queries, relational data
- **AI:** LLM integration, RAG pipelines, agentic AI, structured outputs, model routing
- **Game Development:** Roblox Studio, Luau, client-server architecture, multiplayer systems, gameplay UI, data-driven design
- **Cloud & Infrastructure:** Docker, AWS, Proxmox VE, Nginx, CI/CD, Git
- **Security & Architecture:** API authentication, validation, rate limiting, RBAC concepts, tenant-isolation patterns
- **Frontend:** JavaScript, React, HTML/CSS; working familiarity with TypeScript
- **Languages:** English (fluent), Malay, Mandarin (conversational), German (A2)